

Rahul Bavanandan

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PROFESSIONAL EXPERIENCE

Sr. Machine Learning Engineer | GE Healthcare

Apr 2024 - Present

- Evaluated and implemented new AI and software tools, including **TensorFlow** and **PyTorch**, for applications in healthcare, resulting in a 7% improvement in image analysis accuracy and a 14% reduction in processing time.
- Devised and deployed containerized, serverless machine learning applications on **AWS**, utilizing **EC2**, **ECS**, **S3**, **Lambda**, and other services, to improve scalability and reduce infrastructure costs.
- Built and maintained **CI/CD** pipelines using **Jenkins**, **Terraform**, and **GitHub** to enable automatic testing and deployment of **ML** models.
- Mentored and guided team members on test-driven development (TDD), code refactoring, and coding standards, fostering a culture of collaboration and continuous improvement.

Machine Learning Engineer | Mayo Clinic

Apr 2023 - Mar 2024

- Led design, development, and deployment of **ML** models for healthcare applications, leveraging **deep learning**, **NLP**, and **computer vision** techniques to improve patient outcomes and streamline clinical workflows.
- Collaborated with multidisciplinary teams to analyze vast amounts of healthcare data, including patient records, medical imaging, and genomic information, resulting in a 12% increase in diagnostic accuracy for a specific disease.
- Engineered robust **ML** pipelines and **CI/CD** pipelines to support production processes, ensuring consistent and streamlined development and release of AI software solutions, leading to a 18% reduction in deployment time.
- Offered technical leadership and mentorship to junior engineers, fostering growth in **AI/ML** development and contributing to advancement in healthcare AI.

Data Scientist | Amura Health

Mar 2021 - Jul 2022

- Established and validated CDISC-compliant datasets and analysis results (**TFLs**) using **SAS**, **R**, and **Python**, ensuring data integrity and regulatory compliance for clinical trial data, leading to a 11% reduction in data query rates.
- Collaborated with cross-functional study teams to define data requirements, provide programming support, and ensure timely delivery of high-quality clinical trial data, resulting in an improvement in overall study timelines.
- Mentored 3 junior programmers, offering guidance on programming techniques, data standards, and best practices in clinical trial data management.
- Spearheaded development and implementation of data standards initiatives within the Statistical Data Standards and Analysis (SDSA) department, promoting efficiency and consistency in clinical trial data processing.

Database Analyst | SolvEdge

Apr 2019 - Feb 2021

- Optimized **SQL Server** (2008 R2, 2017, **AWS - RDS**) database systems to enhance query performance by 17% and reduce storage costs by 6% through data compression, ensuring efficient access to critical patient data.
- Implemented data quality checks and validation rules, leading to a 13% reduction in data entry errors and safeguarding reliability of sensitive patient information used in healthcare applications.
- Constructed and maintained over 60 stored procedures and functions to integrate data processing tasks related to patient records, appointments, and billing, thereby enhancing application performance and operational efficiency.
- Migrated **SQL Server** databases to the **AWS RDS** cloud environment, minimizing downtime for healthcare systems, ensuring **HIPAA** compliance and data security through automated backup and recovery processes.

TECHNICAL SKILLS

Programming Languages

: Python, SQL, R, SAS, JavaScript, .NET, PHP, C, C++, Java

Databases & Data Warehousing

: MS SQL Server, PostgreSQL, Snowflake, Star Schema Design, Oracle, DB2

Data Analysis & Visualization

: Power BI, Tableau, QuickSight, Exploratory Data Analysis, Statistical Modeling

Cloud Computing & Big Data

: AWS (IAM, SQS, VPC), Azure, GCP, Hadoop, Hive, Spark

Machine Learning & AI

: Supervised & Unsupervised Learning, Deep Learning, NLP

Data Engineering & Pipelines

: Apache Airflow, ETL Process, Jenkins, Terraform, GitHub, Docker

Other Tools & Technologies

: CI/CD, TDD, Agile Methodologies, SDLC, Jira, Git, Confluence

EDUCATION

Boston University | M.S in Computer Information Systems

Jan 2024

Sastra University | B.Tech in Biotechnology

Oct 2020